



MA 650: Mathematical Models for Infectious Diseases

Course Code	MA 650
Course Name	Mathematical Models for Infectious Diseases
L-T-P-C	3-0-0-3
Prerequisites	None
Intended for	P.G. students
Distribution	Elective
Approval	2nd Adhoc Senate (May 2012)

Course Contents

Over the last two decades, mathematical models have seen a huge development in all aspects of infectious diseases, from microbiology to epidemiology and evolution. The programme covers introductory and advanced concepts in mathematical modelling of infectious diseases, including:

- Mathematical review (calculus, probabilities...)
- Deterministic and stochastic models
- Network analysis
- Within-host dynamics of viral and bacterial infections
- Applied programming with R
- Statistical data modelling
- Computer-based simulations

Text and Reference Books

1. O. Diekmann, and J. A. P. Heesterbeek, **Mathematical Epidemiology of Infectious Diseases: Model Building, Analysis and Interpretation**, Wiley Series in Mathematical & Computational Biology, 2000.
2. D. J. Daley and, J. Gani, **Epidemic Modelling: An Introduction**, Cambridge Studies in Mathematical Biology, 2001.